Phase_C_Information_Systems_Architecture

TOGAF ADM - Phase C: Information Systems Architecture

> ***Purpose***: *Develop Data Architecture and Application Architecture*

Overview

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│                    PHASE C: INFORMATION SYSTEMS ARCHITECTURE             │
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│                                                                          │
│   PURPOSE: Define DATA and APPLICATIONS needed to support business      │
│                                                                          │
│   TWO SUB-PHASES:                                                       │
│   • Data Architecture (what data do we need?)                           │
│   • Application Architecture (what applications do we need?)            │
│                                                                          │
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Key Activities

1. Develop Data Architecture

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│                    DATA ARCHITECTURE                                     │
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│                                                                          │
│   WHAT IT DEFINES:                                                      │
│   • What data entities exist                                            │
│   • How data is structured (logical models)                             │
│   • Where data is stored (physical models)                              │
│   • How data flows between systems                                      │
│   • Data ownership and governance                                       │
│                                                                          │
│   KEY ARTIFACTS:                                                        │
```

```
│    • Conceptual Data Model                                         │
│    • Logical Data Model                                            │
│    • Physical Data Model                                           │
│    • Data Flow Diagrams                                            │
│    • Data Catalog                                                  │
│                                                                     │
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2. Develop Application Architecture

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│                    APPLICATION ARCHITECTURE                         │
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│                                                                     │
│   WHAT IT DEFINES:                                                 │
│    • What applications are needed                                  │
│    • How applications interact                                     │
│    • Application boundaries                                        │
│    • Integration patterns                                          │
│    • Application lifecycle                                         │
│                                                                     │
│   KEY ARTIFACTS:                                                   │
│    • Application Portfolio Catalog                                 │
│    • Application Communication Diagram                             │
│    • Application/Function Matrix                                   │
│    • Interface Catalog                                             │
│                                                                     │
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3. Perform Gap Analysis

- Compare Baseline vs Target for both Data and Application
- Identify what needs to be built, changed, or retired

Example: Gaming Platform

Data Architecture

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DATA ARCHITECTURE:
├── Player Data (profiles, preferences, history)
├── Game Data (scores, achievements, leaderboards)
├── Transaction Data (purchases, wallets, payouts)
└── Analytics Data (behavior, engagement metrics)
```

Application Architecture

```
APPLICATION ARCHITECTURE:
├── Player Management System
├── Game Engine Integration
├── Payment Gateway
├── Bonus & Promotion Engine
├── Responsible Gaming Module
├── Reporting & Analytics Platform
└── Back Office Admin Portal
```

Gap Analysis

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GAP ANALYSIS:
• Current: Separate DBs per game → Target: Unified Player 360 view
• Current: Batch reporting → Target: Real-time analytics
• Current: Point-to-point integrations → Target: Event-driven (Kafka)
```

Data vs Application Architecture

Data Architecture	Application Architecture
WHAT data exists	WHAT applications exist
HOW data is structured	HOW applications interact
WHERE data is stored	WHERE applications run
Data models, schemas	Application portfolio
Data flows	Integration patterns

Key Deliverables

Deliverable	Description
Data Architecture Document	Complete data architecture
Application Architecture Document	Complete application architecture
Gap Analysis Results	What needs to change
Data Models	Conceptual, Logical, Physical
Application Portfolio	List of all applications
Interface Catalog	All integrations documented

Summary

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│                                                                          │
│   PHASE C = "WHAT DATA AND APPS DO WE NEED?"                             │
│                                                                          │
│   DATA ARCHITECTURE:                                                     │
│   • What data entities?                                                  │
│   • How is data structured?                                              │
│   • Where is data stored?                                                │
│                                                                          │
│   APPLICATION ARCHITECTURE:                                              │
│   • What applications?                                                   │
│   • How do they interact?                                                │
│   • What integrations?                                                   │
│                                                                          │
│   OUTPUT: Data Architecture + Application Architecture + Gap Analysis    │
│                                                                          │
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